



YASTEEL RAMPHAL

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DRIVERS LICENSE: Code B

PROFESSIONAL SUMMARY

Final-year Electrical and Information Engineering student with proven experience in industrial automation, data science, and machine learning. Demonstrated expertise in SCADA systems, manufacturing process optimisation, and predictive modelling through internships at SAB-AB InBev and Wits Business Intelligence. Strong technical foundation in control systems, software development, and cross-functional team leadership, seeking to contribute innovative engineering solutions in a dynamic industrial environment.

QUALIFICATIONS

Bachelor's degree: **Final Year**
BSc (Eng) in Electrical and Information Engineering
UNIVERSITY OF WITWATERSRAND

Schooling:

Independent Examinations Board (IEB) Level:
SHANGRI-LA ACADEMY INDEPENDENT SCHOOL

COMMUNITY SERVICE

- SPCA
- Volunteered at old age homes.
- Volunteer at the Food for Life Programme.

WORK EXPERIENCE

South African Breweries (SAB - AB InBev) | Chamdor
Electrical and Information Engineering Intern

Dec – Mar 2025

Worked with Process and Control Engineering and Technical Services team to assist site with the following:

- Developed and maintained SCADA systems for brewery operations, improving process efficiency.
- Learning and applying control engineering concepts to optimise manufacturing processes in FMCG.
- Analysing and troubleshooting Schneider and Krones systems for efficiency improvements.
- Familiarising with industrial instrumentation, including valves, heat exchangers, and pumps.
- Collaborating with cross-functional teams to enhance system performance.
- Designing and commissioning an electrical clean room to ISO standards

UVI Tech | Pomona
Lab Technician

Jan – Oct 2024

- Oversee manufacturing and assist with sales by managing client requests.
- Conducted quality control testing and analysis of pigments and inks.
- Recorded and reported results on manufactured goods.
- Managed deliveries across Gauteng, ensuring timely and safe transport.

Courses:

HARVARDX (edX): 2020

Cloud Computing

Cloud service and deployment models, architecture, security and new technologies

Data Science Program Online

R programming, data wrangling and visualisation, file organisation and reproducible document preparation.

AWS Machine Learning Cloud Camp 2024

Utilising AWS Rekognition and Textract through an API to implement image recognition and text recognition on a locally hosted website.

SELECTED ACHIEVEMENTS

- Awarded the Van Der Linder Trophy for the top achiever in physical science.
- Qualified for the Math Olympiad
- Achieved red belt in Kung Fu

Wits Business Intelligence Services | Johannesburg

Data Science

Oct - Dec 2023

Data science intern responsible for creating and training ML models to create predictive models to support BI. Responsibilities include:

- Performing Exploratory Data Analysis (EDA)
- Preprocessing and cleaning Data
- Machine learning model training
- Model validation
- Implementing a Natural language processing (NLP) model
- API implementation and integration
- Managing and coordinating a team

Wits Business Intelligence Services | Johannesburg

Data Engineering

Jun - Sep 2023

Data engineering intern responsible for managing and creating data pipelines to support Business Intelligence. Responsibilities include:

- Leveraging various schemas to design data warehouses and data lakes
- creating ETL and ELT processes
- Feature engineering
- Cleaning and preparing data
- Integrating cloud services,
- Querying data and representing data for business insights.
- Team management

ACADEMIC PROJECTS *(more projects, details and resources available at: <https://yasteelr.github.io/YasteelR/>)*

EMG Measurement System: Designed a 6-channel electromyographic measurement system for prosthetic control applications.

- System achieves a 120dB CMRR and 1.13% accuracy through precision analogue design
- Implemented real-time digital signal processing for MUAP detection
- Developed complete system from sensor interface to microcontroller integration

Custom Network Synchronisation Protocol:

Developed a custom latency compensation protocol for balancing variation in different client latencies.

- Implemented request-response measurement and time-delay synchronisation techniques
- Achieved 90%+ improvement in timing alignment across varying network conditions

Titanic ML Competition: Implemented ensemble machine learning models for passenger survival prediction.

- Developed family-related feature engineering and RFECV pipeline achieving a 79.5% accuracy
- Ranked 1180 on Kaggle leaderboard among thousands of participants

Open-Source LLM Security Evaluation: Evaluated open-source LLM's detection capability in finding security vulnerabilities and functional correctness.

- Tested open-source models using OWASP and CWE vulnerability datasets for comprehensive security analysis
- Implemented different prompting techniques for comparative performance analysis.
- Demonstrated significant improvements in vulnerability detection accuracy for real world use

SKILLS:

- Electronic Systems Design
- Control System Design
- Krones and Schneider Systems
- Manufacturing Systems Integration
- Industrial Process Optimisation
- Autocad and 3D Modelling
- Software Development
- DevOps and Automation
- C++/Python/SQL /HTML/CSS/JS
- Git/ GitHub and Cmake
- Data Analytics
- Unit Testing And OOP
- Wireless Communication and Signal Modulation
- MATLAB and Simulink
- Technical Lab Skills
- Power BI
- Computer Cabling and Hardware Trouble Shooting
- Logistics and sales
- Machine Learning Model Development
- API Development & Integration
- Data Pipeline Architecture (ETL/ELT)

References

Tafadzwa Raiva | SAB

Technical Services Manager

074 532 5400

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Sabelo Mazibuko | SAB

Manufacturing Systems Manager

076 986 9104

Ashlen Ramsaroop | SAB

Process and Control Engineer

079 693 0441

Joe | UVI Tech

Owner

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